Lecture 7: Markov Chain Monte Carlo

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Announcements

- Reading: Chapter 10 + 11

Markov Chain Monte Carlo

- We want ~~independent~~ iid random sa mples, $\theta^{(s)}$ from $p(\theta \mid y_1, \ldots y_n)$
- Generally hard to find efficient Monte Carlo methods with independent samples (Rej. Sampler, Importance Sampler)
- Alternative, create a sequence of **correlated** samples that converge to the posterior distribution

Efficiency cost.

Discrete-state Markov Chains

Discrete Values

- Let $\theta^{(t)} \in 1, 2, \ldots M$ be the state space for the stationary Mark
- A sequence is called a markov chain if
  $Pr(\theta^{(t+1)} = j \mid \theta^{(t)} = i, \theta^{(t-1)} = i_{t-1} \ldots \theta^{(1)} = i_1) = Pr(\theta^{(t}$
  for all $t \geq 0$

  entire past

  $= P(\theta^{t+1} \mid \theta^{t})$
- $q_{ij} = Pr(\theta^{(t+1)} \mid \theta^{(t)})$ is the transition probability from state i t
- The *Markov property*: given the entire past history, $\theta^{(1)}, \ldots \theta^{(t)}$, t
  recent $\theta^{(t+1)}$ depends only on the immediate past, $\theta^{(t)}$
- The $M \times M$ matrix $Q = (q_{ij})$ is called the *transition matrix* of the Markov Chain

The Transition Matrix (Example)

- The $M \times M$ matrix $Q = (q_{ij})$ is called the *transition matrix* of the Markov Chain

3-state example

$$Q = \begin{array}{cc} & \text{To} \\ \text{From} & \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \end{array}$$

sum to 1.

$$(q_{ij}) = Pr(\text{go to } j \mid \text{in state } i)$$

- The rows of the transition matrix sum to 1
- Note: $Q^n = (q_{ij}^{(n)})$ is is the probability of transitionining from i to j in n steps
- A Markov Chain is called *regular* if for some n, $Q_{ij}^n > 0$ for all i, j). E.g.it is possible to go from i to j in a finite number of step).

Matrix power n.

The limiting distribution

- *The limiting distribution* is $\pi_j = lim_{n \to \infty} Q_{ij}^n > 0$ independent of the starting point i.
- A regular Markov chain has a *limiting probability distribution.
- The limiting distribution of a regular transition matrix is the unique nonnegative solution to the equation $\pi Q = \pi$
  - This is called the *stationary distribution* / π
  - ~~Does depend on where the chain starts~~

Markov Chain Example

- Sociologists often study social mobility using a Markov chain.
- In this example, the state space if {low income, middle income, and high income} of families
- Let $\mathbf{Q}$ be the transition matrix from parents income to childrens income:

Your Income.

Parents

	Lower	Middle	Upper
Lower	0.40	0.50	0.10
Middle	0.05	0.70	0.25
Upper	0.05	0.50	0.45

$= Q$

Multi-step Transition Probabilities

2-step transition probabilities

$$\mathbf{Q}^2 = \mathbf{Q} \times \mathbf{Q} = \begin{array}{rl} & \begin{array}{ccc} & \text{Yours} & \text{Kids} \end{array} \\ \text{Parents} & \begin{vmatrix} 0.1900 & 0.6000 & 0.2100 \\ 0.0675 & 0.6400 & 0.2925 \\ 0.0675 & 0.6000 & 0.3325 \end{vmatrix} \end{array}$$

4-step transition probabilities

$$\mathbf{Q}^4 = \mathbf{Q}^2 \times \mathbf{Q}^2 = \begin{vmatrix} 0.0908 & 0.6240 & 0.2852 \\ 0.0758 & 0.6256 & 0.2986 \\ 0.0758 & 0.6240 & 0.3002 \end{vmatrix}$$

Multi-step Transition Probabilities

4-step transition probabilities

$$\mathbf{Q}^4 = \mathbf{Q}^2 \times \mathbf{Q}^2 = \begin{vmatrix} 0.0908 & 0.6240 & 0.2852 \\ 0.0758 & 0.6256 & 0.2986 \\ 0.0758 & 0.6240 & 0.3002 \end{vmatrix}$$

8-step transition probabilities

$$\mathbf{Q}^8 = \mathbf{Q}^4 \times \mathbf{Q}^4 = \begin{vmatrix} 0.0772 & 0.6250 & 0.2978 \\ 0.0769 & 0.6250 & 0.2981 \\ 0.0769 & 0.6250 & 0.2981 \end{vmatrix}$$

Converging to Stationary Distn.

The stationary distribution

$\pi = (\pi_1, \pi_2, \pi_3)$

$$\mathbf{Q}^\infty = \begin{pmatrix}1\\1\\1\end{pmatrix}\pi = \begin{vmatrix}\pi_1 & \pi_2 & \pi_3\\ \pi_1 & \pi_2 & \pi_3\\ \pi_1 & \pi_2 & \pi_3\end{vmatrix}$$

$\pi_1 + \pi_2 + \pi_3 = 1$

```
Q <- matrix(c(0.4, 0.05, 0.05,
              0.5, 0.7, 0.5,
              0.1, 0.25, 0.45), ncol=3)
p <- eigen(t(Q))$vectors[, 1]
stationary_probs <- p/sum(p)
stationary_probs
```

```
[1] 0.07692308 0.62500000 0.29807692
```

```
stationary_probs %*% Q
```

```
           [,1]  [,2]      [,3]
[1,] 0.07692308 0.625 0.2980769
```

π stationary $\Longrightarrow$

$$\pi Q = \pi$$

π is an eigenvector of Q, with e-value $= 1$.

2/11

- Hwk due date mistake.
- No Class Monday.
- Project Details Wed.

Markov Chain Monte Carlo

- Incredible idea: create a Markov Chain with the desired limiting distribution
  - Want the limiting distribution to be the posterior distribution $P(\theta | y)$
- Unlike the previous examples, we will mostly work with *infinite* state space
- Want $p(\theta^{(t+1)} \mid \theta^{(t)})$ to have limiting distribution $p(\theta \mid y)$

"transition kernel"

  - If we run the random walk for long enough, $\theta^{(t)}$ will be distributed approximately according to $p(\theta \mid y)$

$$P(\theta_t < k) \xrightarrow{t \to \infty} P(\theta < k \mid y) \quad \forall k$$

Detailed Balance

- Assume a Markov Chain θ_t, with stationary distribution $\pi(s)$, and let

Shorthand

$$p(s, s') = Pr(\theta_{t+1} = s \mid \theta_t = s')$$

The chain is said to be reversible with respect to π or to satisfy detailed balance with respect to π if

$$\boxed{\pi(s')p(s, s') = \pi(s)p(s', s) \quad \forall s, s'}$$

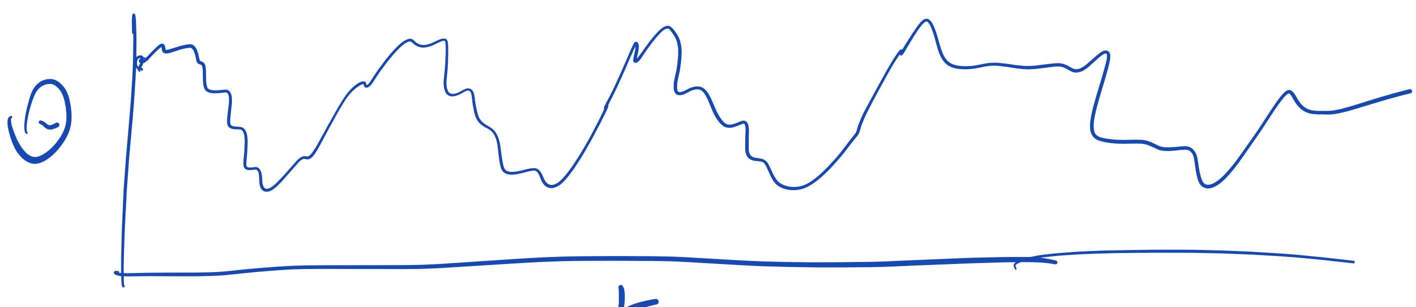
Prob start is s', go to s = Prob. start s, go to s'

Detailed Balance

$\pi = P(\theta | y)$, find $P(\theta_{t+1} | \theta_t)$ that satisfies D.B.

- If $\pi(s)$ and $p(s, s')$ satisfy detailed balance then π is the unique stationary distribution of the Markov Chain.
- Goal: devise a transition kernel $p(s, s')$ for which $\pi(\theta = s \mid y) = \pi(s)$ satisfies detailed balance
- Detailed balance is a stronger condition than required for a stationary distribution

sufficient condition.



Monte Carlo CLT

$\theta^s \overset{iid}{\sim} P(\theta|y)$

- Reminder: $\overline{\theta} = \sum_{s=1}^{S} \theta^{(s)}/S$ and S is the number of samples.
- If the samples are independent,

CLT

$$p(\overline{\theta}) \to N\left(E[\theta|y], \frac{\text{Var}(\theta \mid y_1, \ldots y_n)}{S}\right)$$

- If the samples are from a Markov Chain with θ_1 from the stationary distribution, then

$$p(\overline{\theta}) \to N\left(\theta, \frac{\overline{\sigma}^2}{S}\right)$$

where $\overline{\sigma}^2 = \underbrace{Var(\theta \mid y_1, \ldots y_n)}_{\text{same as with iid}} + \underbrace{2\sum_{s=1}^{\infty} Cov(\theta_1, \theta_s)}_{\text{Penalty from having correlated samples.}}$

$Cov(\theta_1, \theta_s)$ has to decay as s gets big

Goal $E[\theta | y]$. Assume $\underline{\theta_i \sim p(\theta|y)}$

$\theta_1, \ldots \theta_T \sim$ MCMC

$$Var(\bar{\theta}) = Var\left(\frac{1}{T}\sum^{T} \theta_i\right)$$

$$= \frac{1}{T^2} \sum_{i,j} Cov(\theta_i, \theta_j)$$

$$= \frac{1}{T^2}\left(\sum_{i=1}^{T} Var(\theta_i) + \sum_{i}^{T}\sum_{j>i}^{T} 2\, Cov(\theta_i, \theta_j)\right)$$

e.g. $Cov(\theta_1, \theta_k) = Cov(\theta_{10}, \theta_{10+k})$

$$= \frac{1}{T^2}\left(T\, Var(\theta|y) + 2\sum_{k}^{T-1} T(T-k)\, Cov(\theta_1, \theta_{1+k})\right)$$

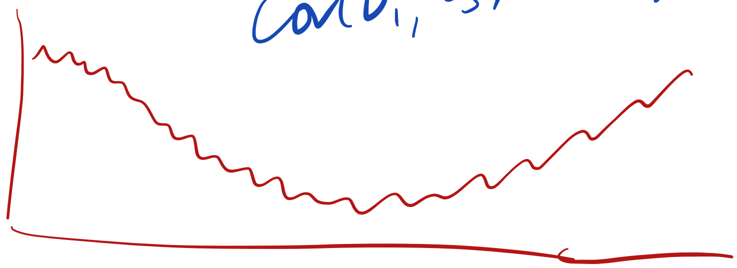
$<<$ $T \to \infty$

$$= \underbrace{\frac{Var(\theta|y)}{T}}_{\text{Independent MC Error}} + 2\sum_{k}^{\infty} Cov(\theta_1, \theta_{1+k})$$

MC 1

$\mathrm{cov}(\theta_1, \theta_5)$ is large

Θ



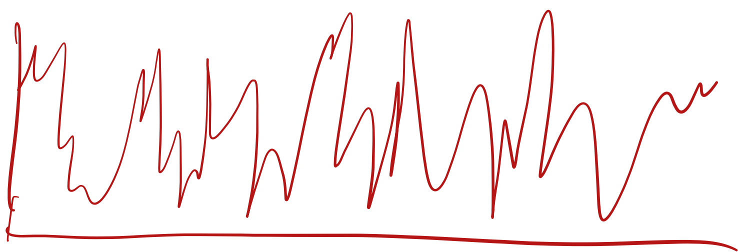
t

Bad! High M.C. error.

MC 2

$\mathrm{cov}(\theta_1, \theta_5)$ is small.

Θ



less correlated, lower Monte Carlo error.

The Independence Sampler

1. Initialize θ_0 to be the starting point
2. Generate the candidate θ^* from a proposal distribution, $J(\theta^*)$ which is indepednent of θ_t
3. Compute $r = \min(1, \frac{p(\theta^*|y)}{p(\theta_t|y)})$
4. Set $\theta_{t+1} \leftarrow \theta^*$ with probability r.

"Acceptance Prob"

Still a Markov Chain "MCMC"

- Generate a uniform random number $u \sim Unif(0, 1)$
- If $u < r$ we accept θ^* as our next sample
- Else $\theta_{t+1} \leftarrow \theta_t$ (we do not update the sample this time)

Keep previous value if θ^* is rejected.

Intuition

3. $r = \min\left(1, \dfrac{P(\theta^* | y)}{P(\theta_t | y)}\right)$, accept w.p. r.

$P(\theta^* | y) > P(\theta_t | y)$ accept!

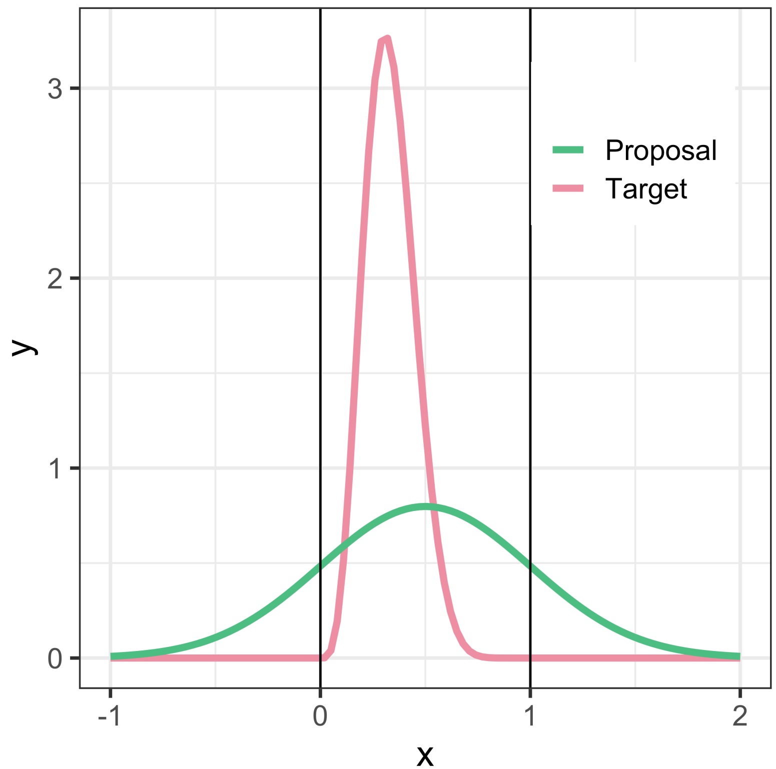
$P(\theta^* | y) < P(\theta_t | y)$, accept w.p. r

Relative Frequency of θ^* vs θ_t

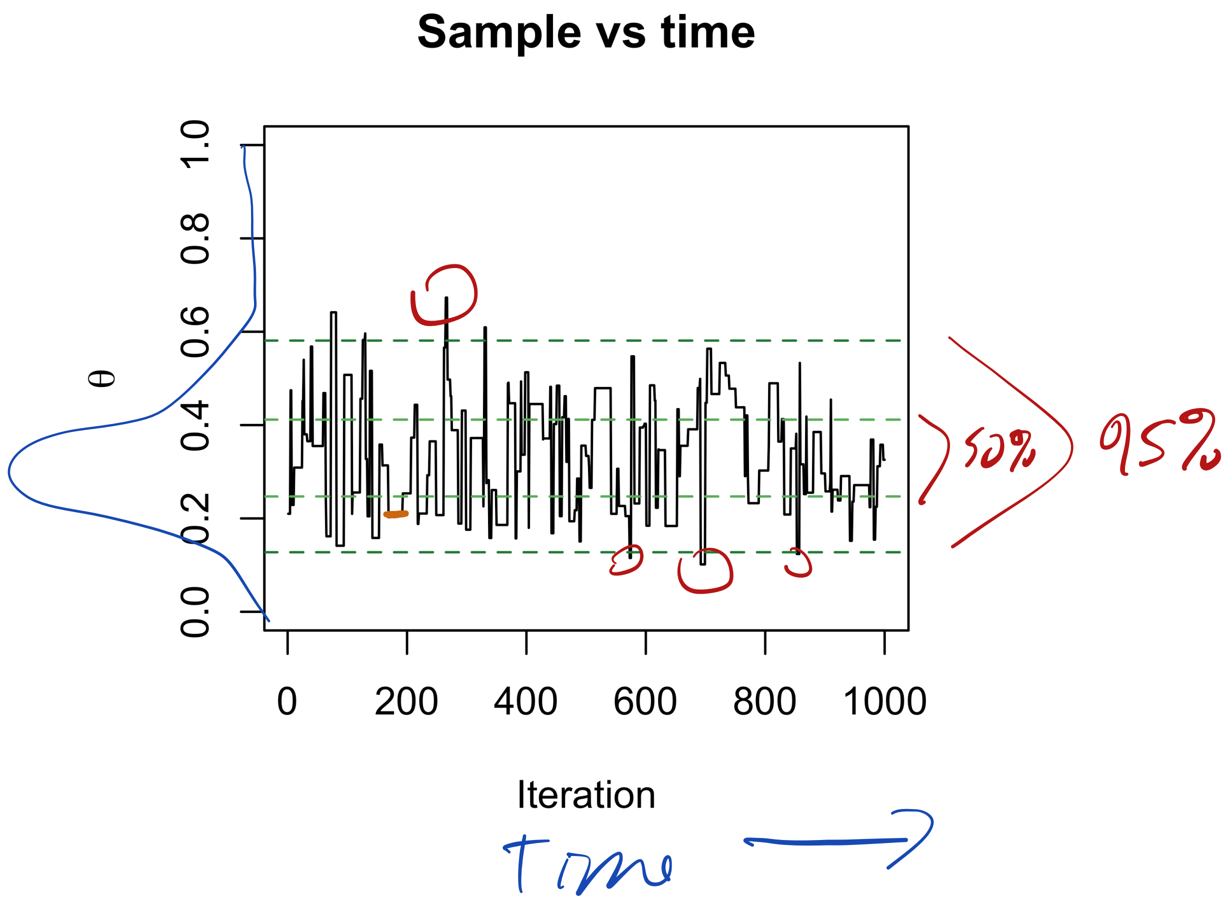
Should be $\dfrac{P(\theta^* | y)}{P(\theta_t | y)}$

An Example

- Let $P(\theta \mid y)$ be a Beta(5, 10) posterior distribution
- Propose from a distribution $J(\theta^*) \sim N(0.5, 1)$

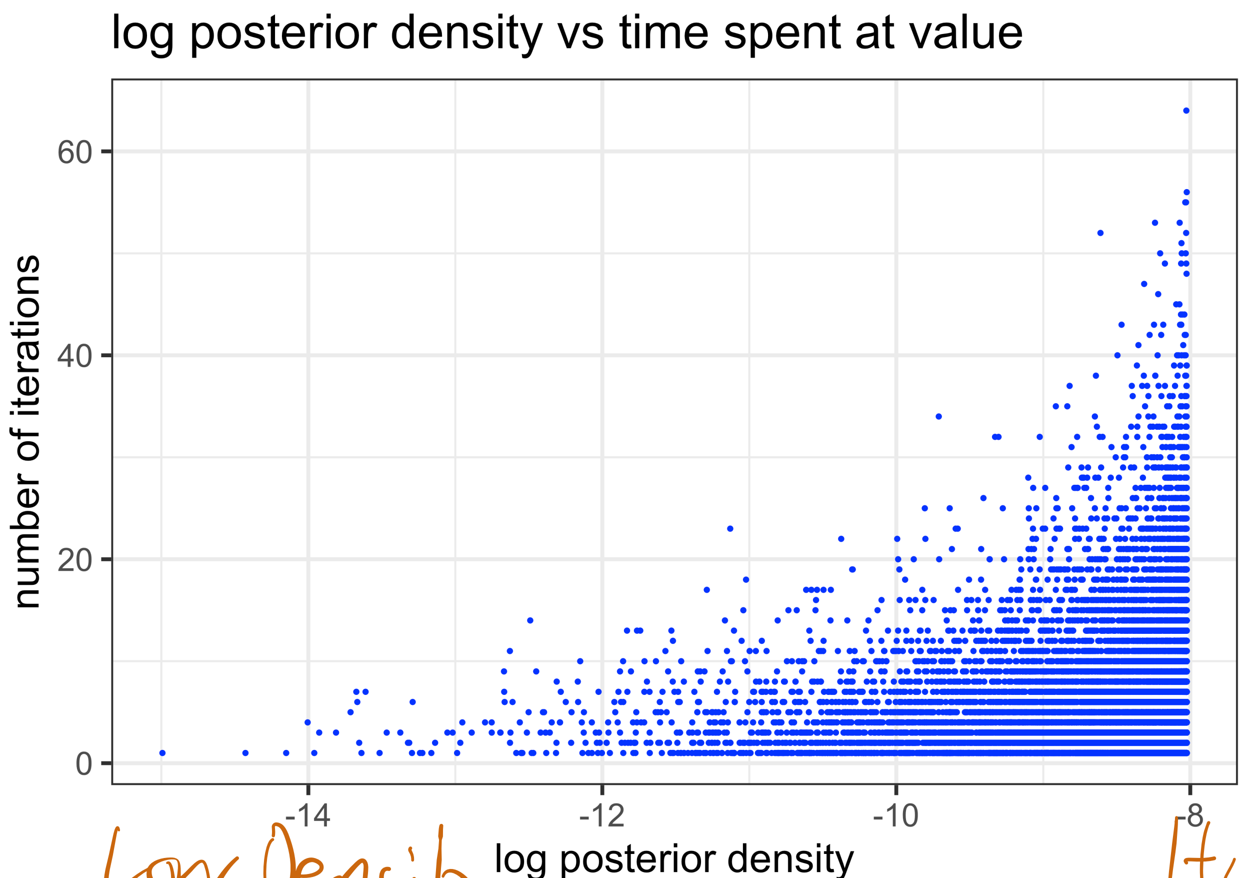


Independence Sampler



Note and source of confusion: samples are correlated over time for the “independence sampler”

Weighting by waiting

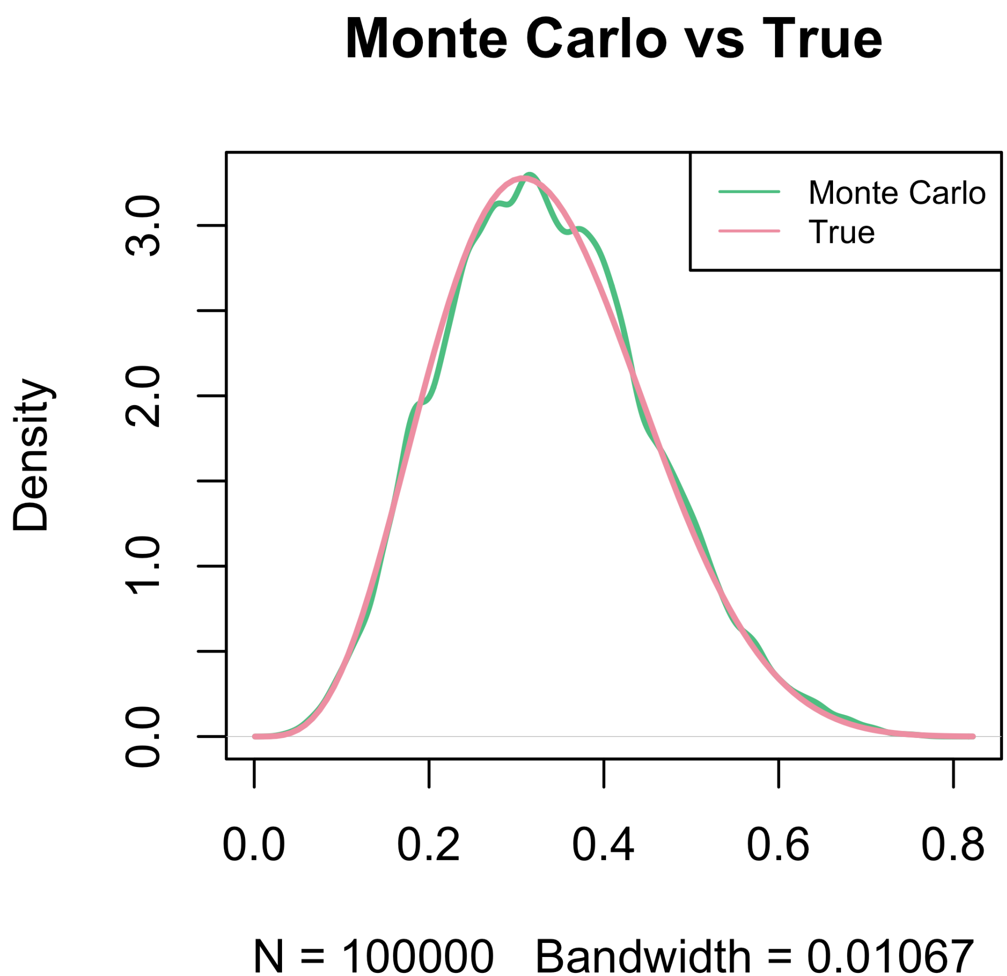


Low Density

High Density

Where did the sampler spend the most time? Where does it quickly leave?

Independence Sampler



Metropolis Algorithm

- Let $p(\theta \mid y)$ be the posterior distribution
- The goal is to define a transition kernel, $p(\theta_{t+1} \mid \theta_t)$ such that $p(\theta \mid y)$ is the stationary distribution for the proposal

The Metropolis Algorithm

- Generalize the independence sampler
  - $J(\theta^* \mid \theta_t) = J(\theta^*)$

Depends on current value of θ_t

- Metropolis: Allow the proposal distribution to depend on the most recent sample
  - $J(\theta^* \mid \theta_t)$, e.g. $\theta^* \sim N(\theta_t, 1)$
- Metropolis sampler: a "moving" proposal distribution

The Metropolis Algorithm

$$\theta^* \sim J(\theta^* | \theta_t)$$

$$\theta_{t+1} \leftarrow \theta^* \quad \text{w.p.} \quad r = \min\left(1, \frac{p(\theta^* | y)}{p(\theta_t | y)}\right)$$

else

$$\theta_{t+1} \leftarrow \theta_t$$

Intuition

If θ^* has higher density, take it.

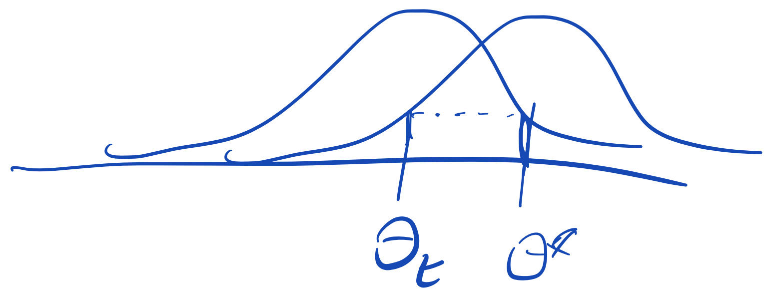
Else take w.p. prop. to density ratio.

For Metropolis, need

$$J(\theta^* | \theta_t) = J(\theta_t | \theta^*)$$

"symmetric proposal"

$J \sim N(\theta_t, 1)$



ratio.

Metropolis-Hastings Algorithm

- The Metropolis-*Hastings* algorithm allows us to use non-symmetric proposals

$J(\theta^* | \theta_t) \neq J(\theta_t | \theta^*)$

- The Hastings correction is needed when $J(\theta^* \mid \theta_t) \neq J(\theta^t \mid \theta^*)$

$$r = \min(1, \frac{p(\theta^* \mid y)}{p(\theta_t \mid y)} \frac{J(\theta^t \mid \theta^*)}{J(\theta^* \mid \theta_t)})$$

Proposal Density.

- For symmetric proposals $\frac{J(\theta^t|\theta^*)}{J(\theta^*|\theta_t)} = 1$

Detailed Balance Proof M-H Satisfies D. B.

Two states, θ_a, θ_b

Show: $P(\theta_a|y)P(\theta_b|\theta_a) = P(\theta_b|y)P(\theta_a|\theta_b)$

Assume without loss of generality

$$P(\theta_b|y)J(\theta_a|\theta_b) \leq P(\theta_a|y)J(\theta_b|\theta_a)$$

Proof:

$$P(\theta_a|y)P(\theta_b|\theta_a) =$$

$$\cancel{P(\theta_a|y)}\underbrace{\cancel{J(\theta_b|\theta_a)}}_{\text{proposal prob}}\underbrace{\frac{P(\theta_b|y)J(\theta_a|\theta_b)}{\cancel{P(\theta_a|y)}\cancel{J(\theta_b|\theta_a)}}}_{\text{acceptance prob "r"}}$$

$$= P(\theta_b|y)J(\theta_a|\theta_b)$$

$$= P(\theta_b|y)P(\theta_a|\theta_b)$$

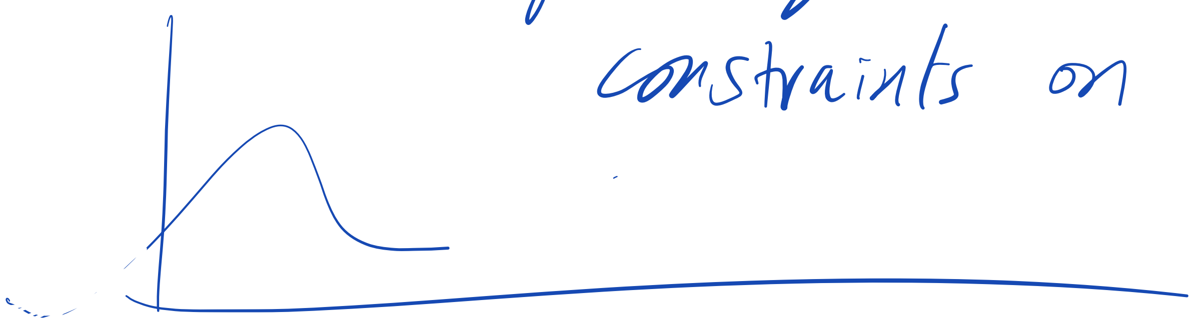
$$P(\theta_a|\theta_b) = J(\theta_a|\theta_b) \times r$$

$r = 1$ B/C

$$P(\theta_b | y) J(\theta_a | \theta_b) = P(\theta_a | y) J(\theta_b | \theta_a)$$

$$r = \frac{\dfrac{P(\theta_a | y) J(\theta_b | \theta_a)}{P(\theta_b | y) J(\theta_a | \theta_b)}}{}$$

Symmetry broken w/ constraints on parameters



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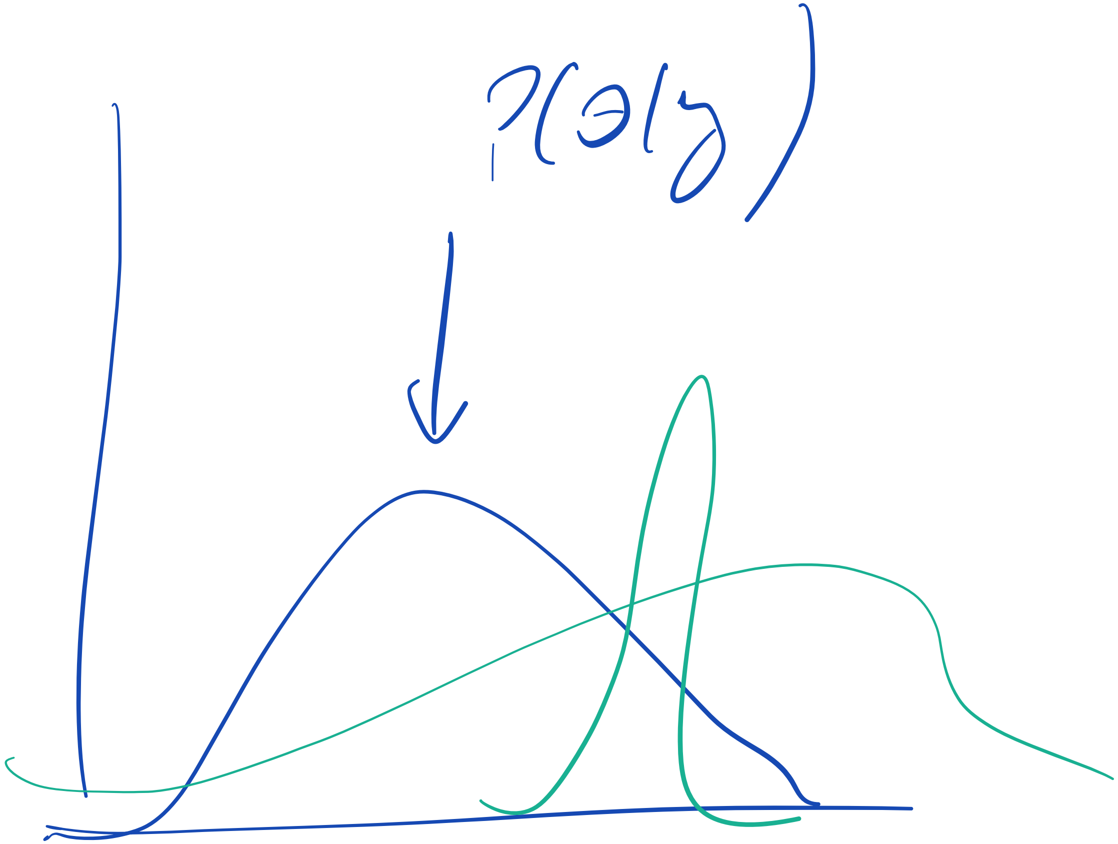
Aside: Arianna Rosenbluth

https://en.wikipedia.org/wiki/Arianna_W._Rosenbluth

An Example



How to choose $J(\theta^* | \theta_t)$, what are the tradeoffs?

Example

$J \sim N(\theta_t, \tau^2)$

What τ^2 do I use?

Small variance proposal

$\tau^2 = 0.01$ ("small" jump)

```r
acf(rw_small, lag=100)
```

```r
print(sprintf("Effective sample size: %.2f, Rejection Rate: %.2f",
              effectiveSize(rw_small), rejectionRate(as.mcmc(rw_small))))
```

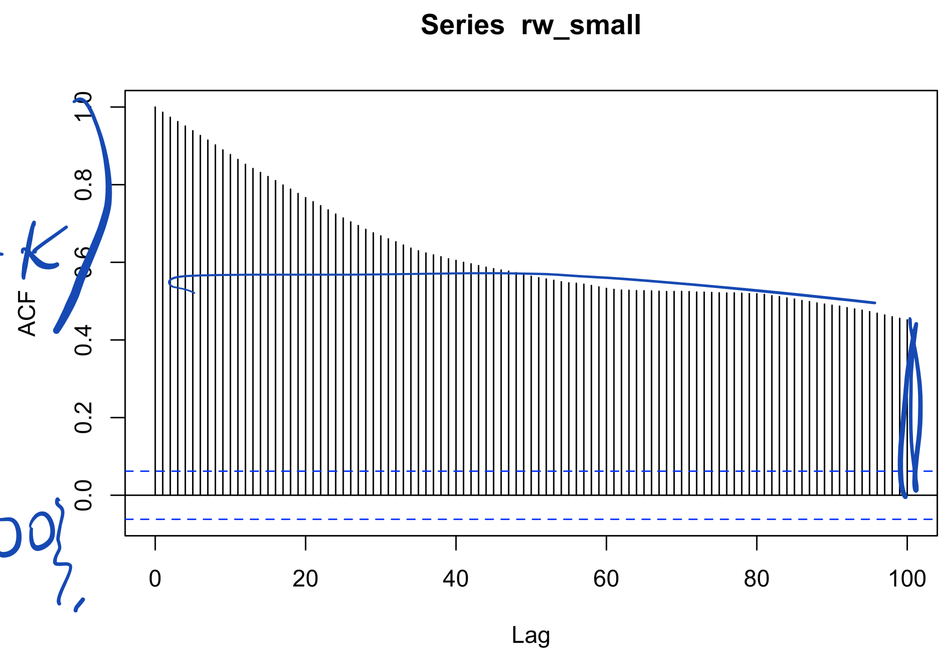
```
[1] "Effective sample size: 5.78, Rejection Rate: 0.04"
```

Plot

$Cov(\theta_t, \theta_{t+k})$

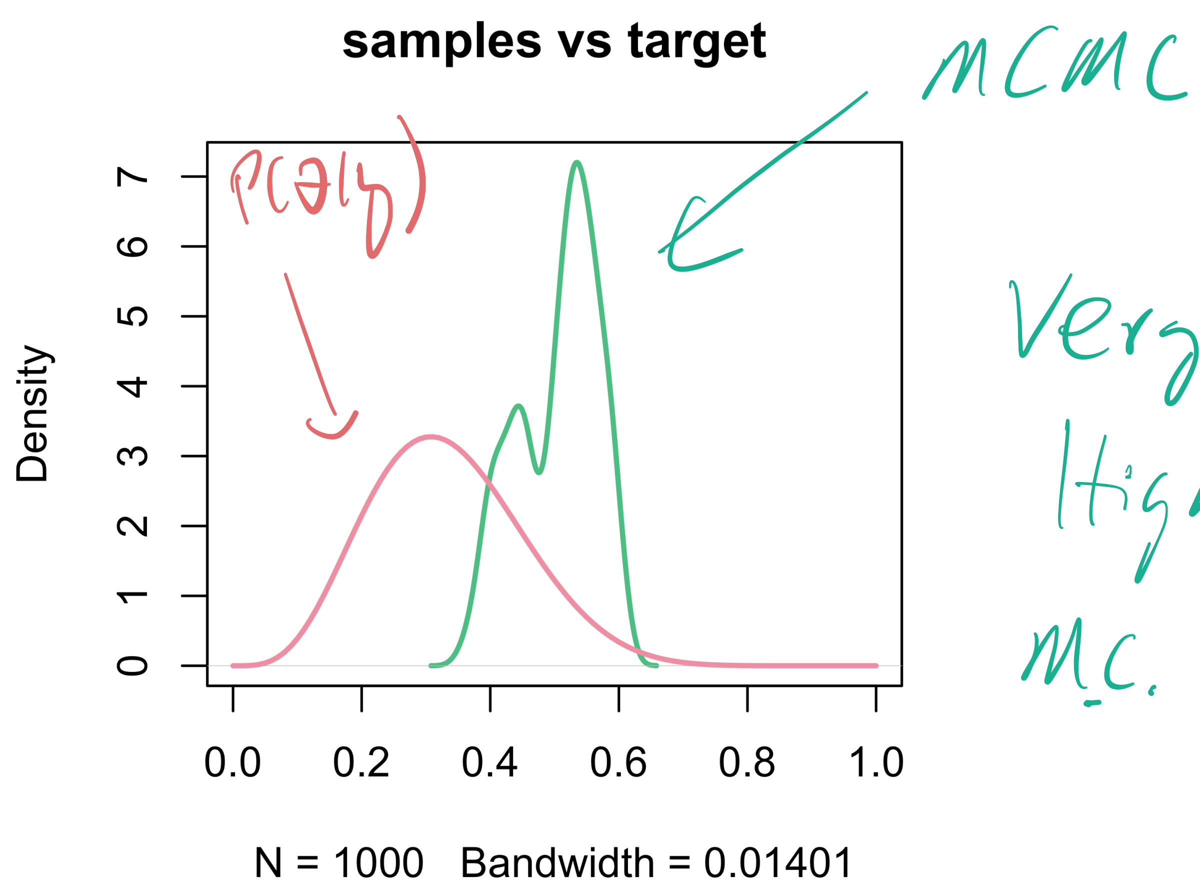
for

$k \in \{1, \ldots, 100\}$,



Small variance proposal

$\tau^2 = 0.01$ ("small" jump)



Very High MC. Error.

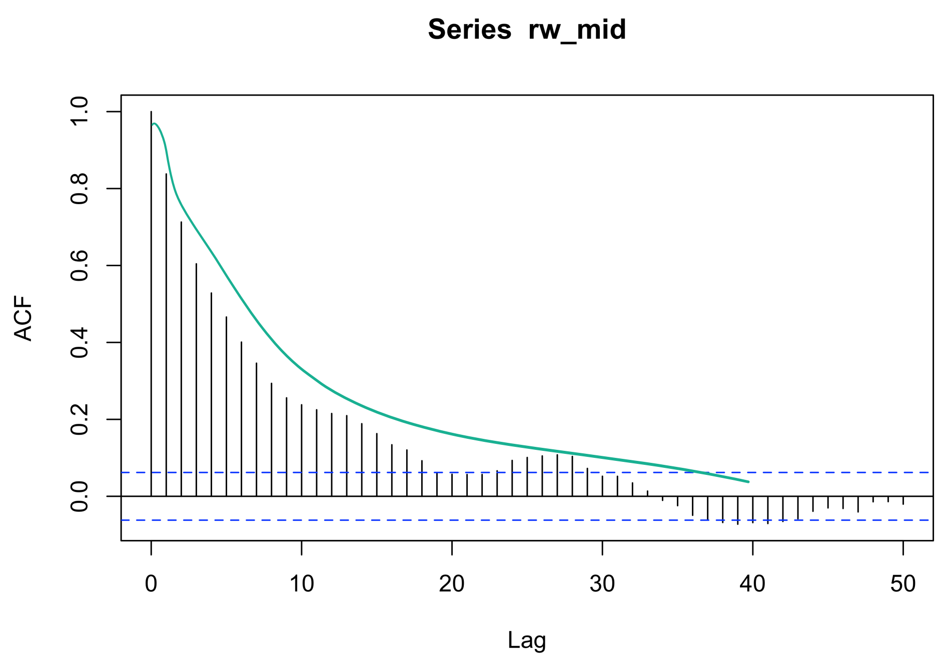
The Metropolis Algorithm

$\tau^2 = 0.1$ ("medium" proposal variance)

```r
acf(rw_mid, lag=50)
```

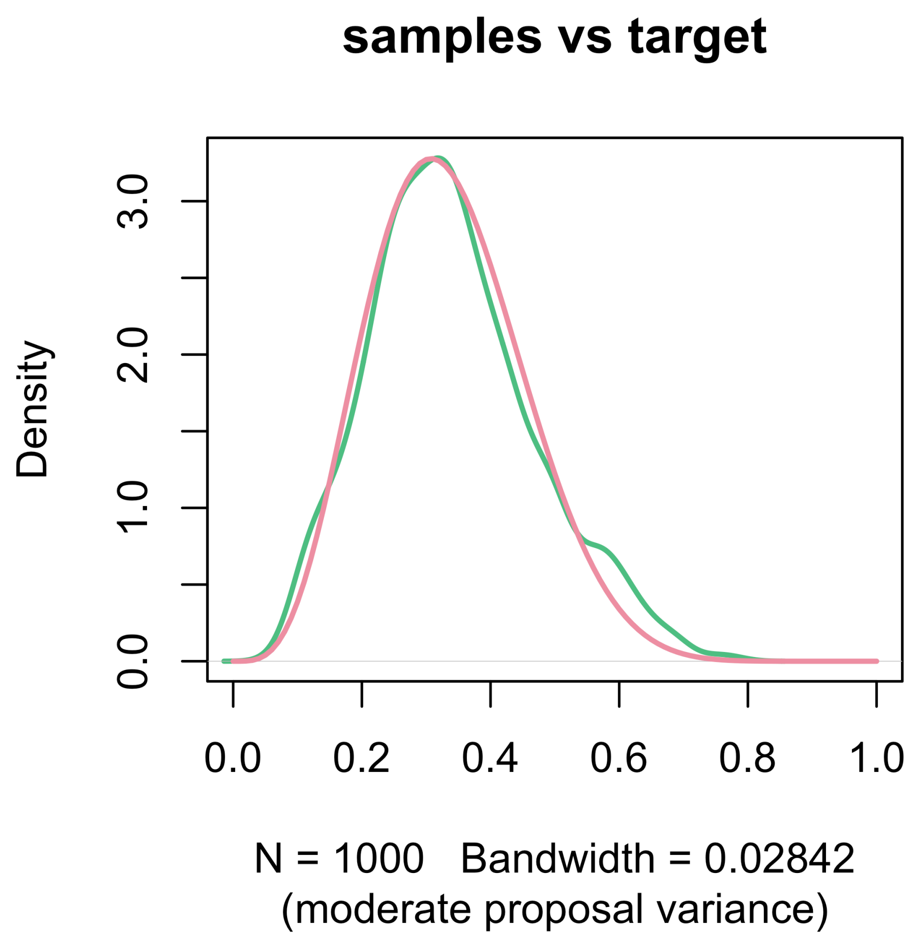
```r
print(sprintf("Effective sample size: %.2f, Rejection Rate: %.2f",
              effectiveSize(rw_mid), rejectionRate(as.mcmc(rw_mid))))
```

```
[1] "Effective sample size: 88.09, Rejection Rate: 0.26"
```



The Metropolis Algorithm

$\tau^2 = 0.1$ ("medium" proposal variance)



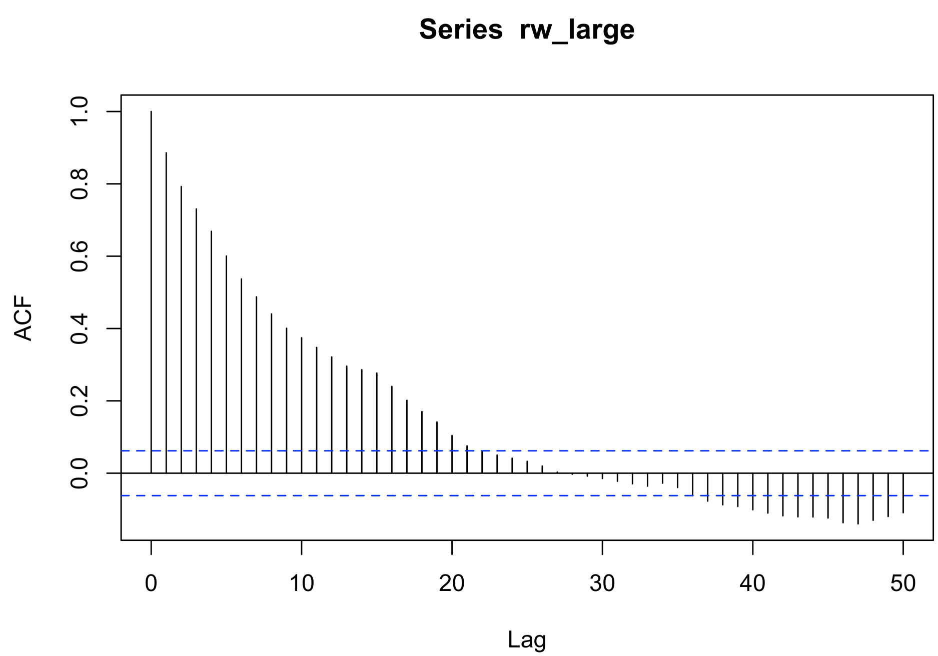
The Metropolis Algorithm

$\tau^2 = 2$ ("large" jump)

```
acf(rw_large, lag=50)
```

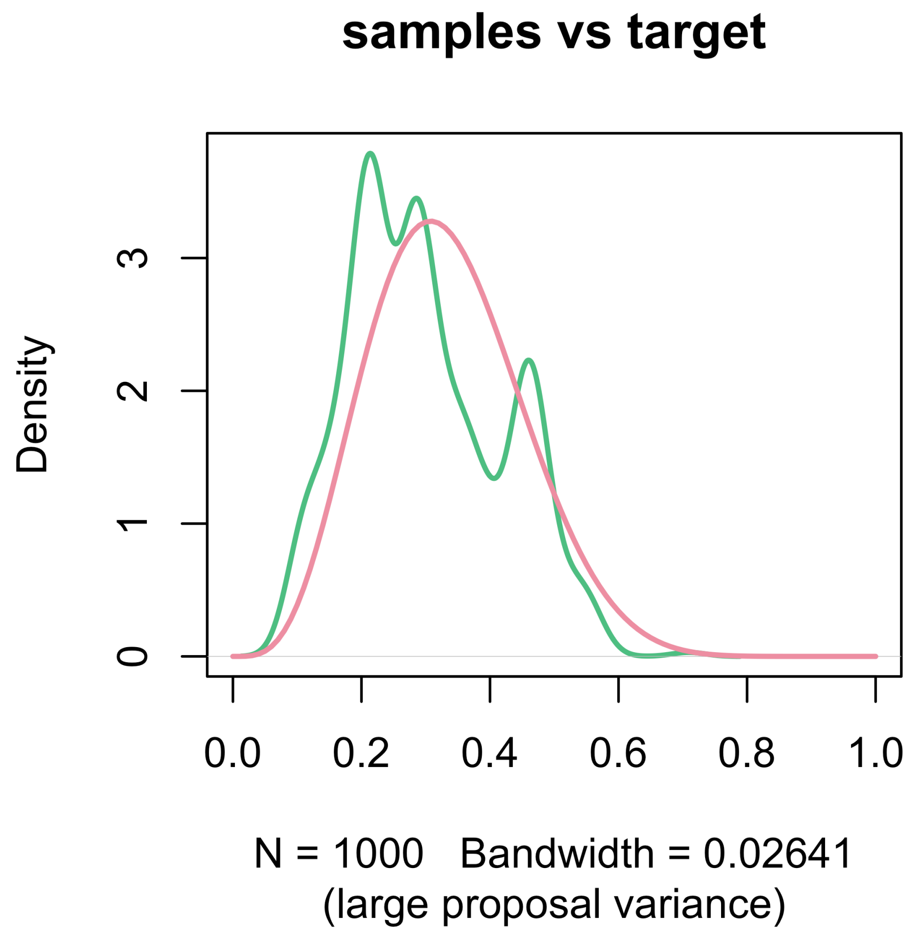
```
print(sprintf("Effective sample size: %.2f, Rejection Rate: %.2f",
              effectiveSize(rw_large), rejectionRate(as.mcmc(rw_large))))
```

```
[1] "Effective sample size: 45.81, Rejection Rate: 0.92"
```



The Metropolis Algorithm

$\tau^2 = 2$ ("large" proposal variance)



Small, Moderate and Large Proposal variance

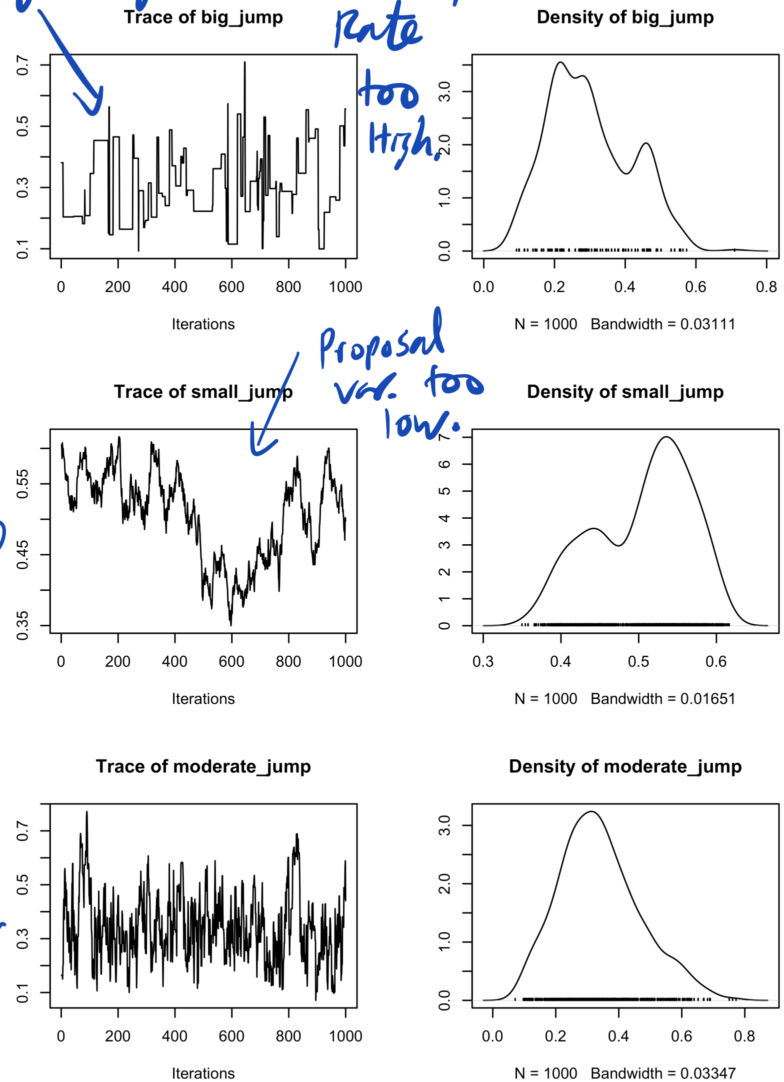
"City Scape" Rejection Rate too High.

τ^2 large

Proposal var. too low.

τ^2 small

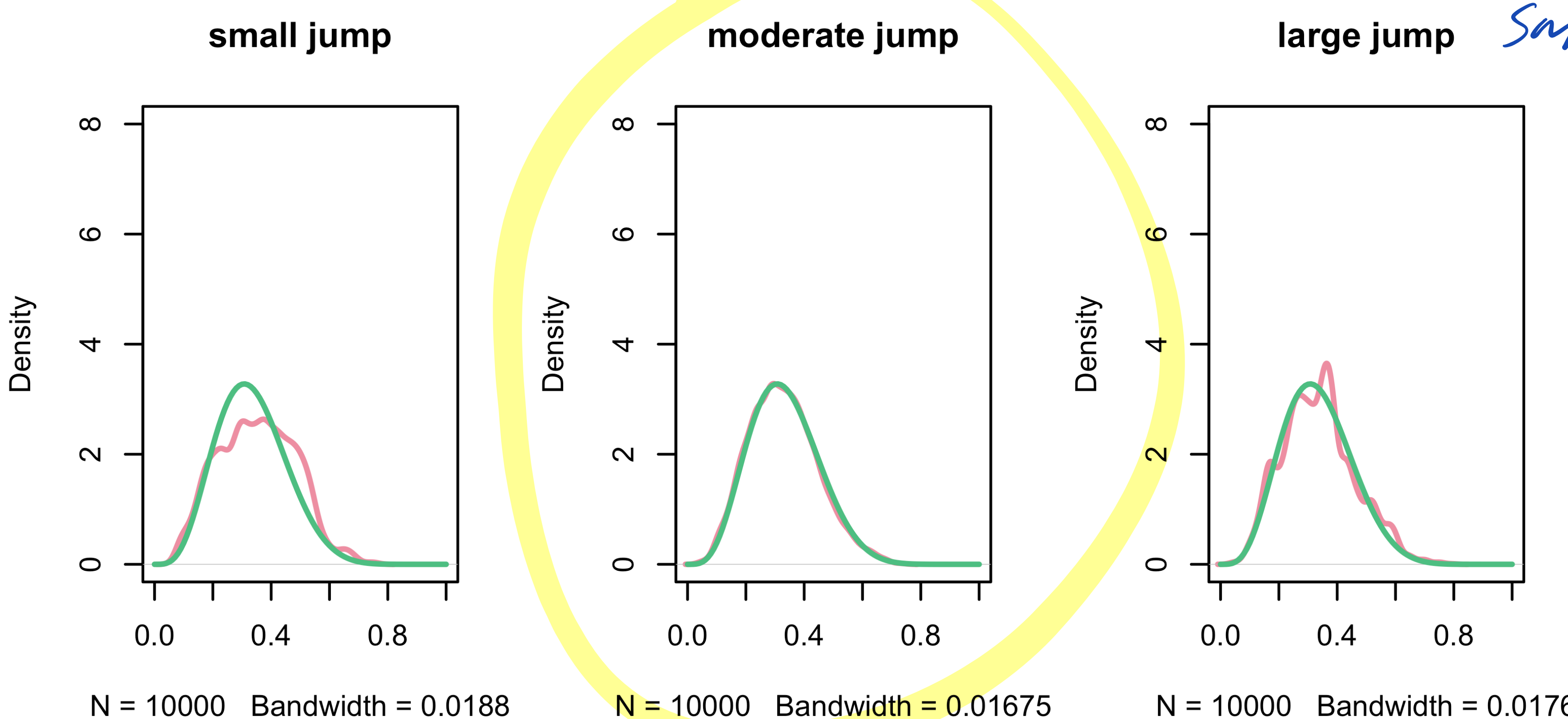
Goldilocks



10,000 Samples

```
[1] "Effective sample size (small proposal variance): 14.36"
[1] "Effective sample size (medium proposal variance): 1017.53"
[1] "Effective sample size (large proposal variance): 483.50"
```

As good as 1000 indep samples.



Effect Sample Size

$$\text{M.C. Error} = \frac{\bar{\sigma}^2}{S} = \frac{Var(\theta|y) + 2\sum Cor(\theta_1, \theta_{1+K})}{S}$$

$$\frac{\bar{\sigma}^2}{Var(\theta|y)} = 1 + 2\frac{\sum_{K}^{\infty} Cor(\theta_1, \theta_K)}{Var(\theta|y)}$$

$$Cor(\theta_1, \theta_K) = Var(\theta|y)\,\rho_K \leftarrow Cor(\theta_1, \theta_K)$$

$$= 1 + 2\sum^{\infty} \rho_k$$

MC error for indep = MC error for dep.
solve for n_{eff}.

$$n_{eff} = \frac{n_{iters}}{1 + 2\sum^{\infty} \rho_k}$$

How long you ran the MCMC

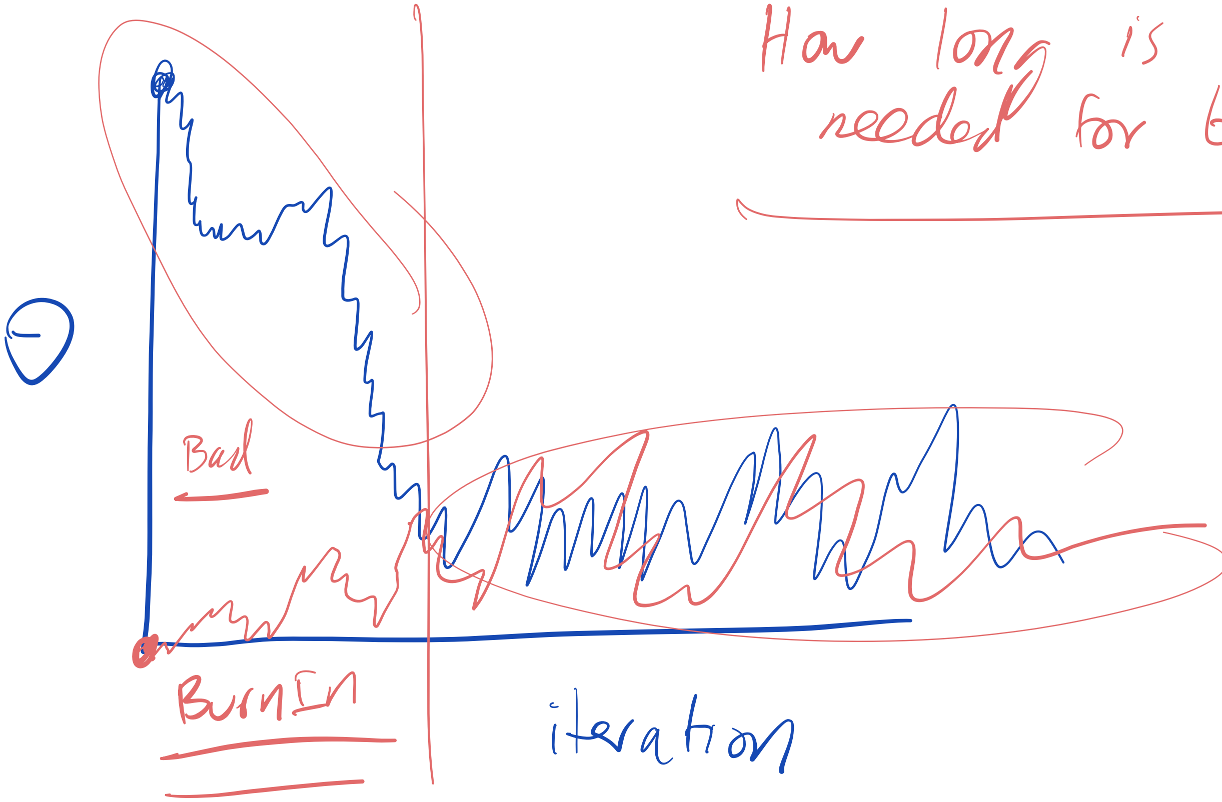
$$\frac{\mathrm{var}(\theta \mid y)}{n_{eff}} = \frac{\sigma^2}{n_{iters}}$$

solve

independent draws

error for dependent.

How long is needed for burnin?



Compare samples from multiple chains.

MCMC diagnostics

- Diagnose the chain(s) performance by examining:
  - Rejection rate Not too low or high
  - Autocorrelation Want as close to 0 as possible.
  - Effective sample size Large as possible.
  - Rhat

MCMC diagnostics

- **Rejection rate**: if rejection rate is high, the proposal density is proposing “too far away” from the current sample
- Means we are often keeping the previous sample
- Sampler is “sticky”. Traceplots look like cityscapes.

MCMC diagnostics

- **Autocorrelation**: if samples are highly correlated, the proposal density is proposing “too close” to the current sample
  - Highly correlated implies the Markov chain is mixing slowly
- The mixing time of a Markov chain is the time until the Markov chain is “close” to its limiting distribution.

MCMC diagnostics

- **Effective sample size:** correlated chain of samples is equivalent to this number of independent samples
  - High rejection rate implies a lot of duplicate samples so effective size is smaller than number of iterations
  - High autocorrelation means neighboring samples are very similar (even if not exactly the same)

Reminder: $\overline{\sigma}^2 = Var(\theta \mid y_1, \ldots y_n) + 2\sum_{s=1}^{\infty} Cov(\theta_1, \theta_s)$

$$n_{\text{eff}} = \frac{mn}{1+2\sum_{t=1}^{\infty}\rho_t}$$

Metropolis Algorithm

- In the Beta example, the accept ratio, $\min(1, \frac{p(\theta^*|y)}{p(\theta_t|y)})$ is zero when $\theta^* > 1$ or $\theta^* < 0$
- If τ^2 (proposal variance) too large, you will often reject your proposal
  - Proposing far from your current location may move you too far out of the high density areas
  - This makes for a “sticky” chain (stay at current sample for a long time)
- τ^2 too small, the chain explore the parameter space slowly

Rhat

- Important to assess convergence by checking whether the chains "mix"
- Visual check: look at traceplots of multi-chains
- Quantitative check: look at within- and between- chain variation

Numerical Summary.

Rhat

within
Chain
Avg.

Overall
Average
(all chains)

- Assume m chains and n draws per chain

Want B small.

- Let $B = \frac{1}{m-1}\sum_{j=1}^{m}\left(\bar{\theta}_{.j} - \bar{\theta}_{..}\right)^2$ be the between chain variance
- Let $W = \frac{1}{m}\sum_{j=1}^{m} s_j^2$ be the average within chain variance

Then we define $\widehat{\text{var}}^+(\psi \mid y) = \frac{n-1}{n}W + B$ and the Rhat as

$$\widehat{R} = \sqrt{\frac{\widehat{\text{var}}^+(\psi|y)}{W}} = \sqrt{1 + B/W}$$

Rhat ≈ 1, < 1.05 ok.

Rhat is very large.